PART 9 · WORKFLOW

From unrigged model to Track It asset

A step-by-step walkthrough of taking a plain, unrigged model (here a **four-wheel mine cart**) and turning it into a library vehicle you can drop onto any track, couple into a train, and animate. The steps marked Tool are proposed add-on features; everything else can be done by hand today.

9.1 The whole journey

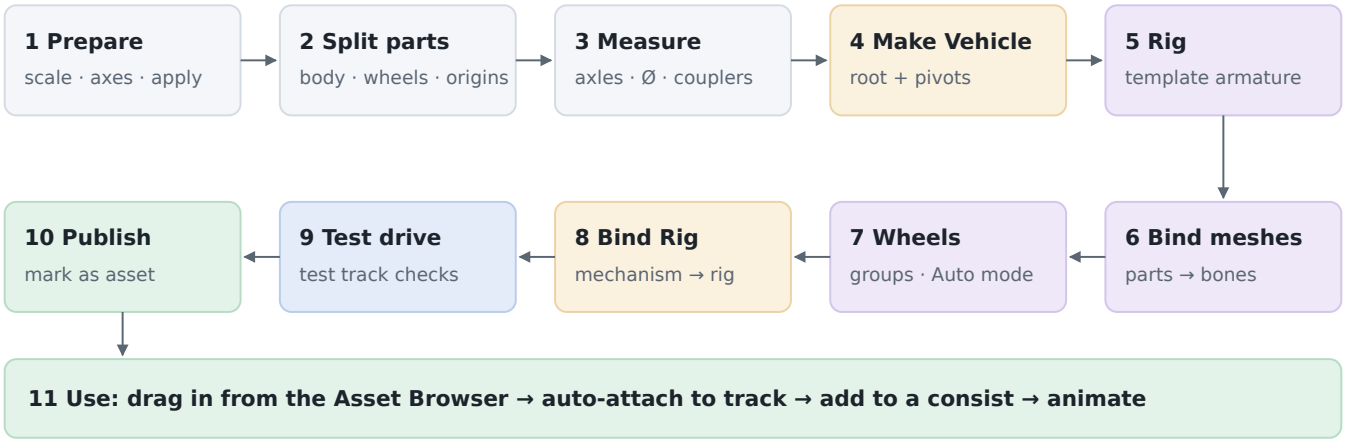
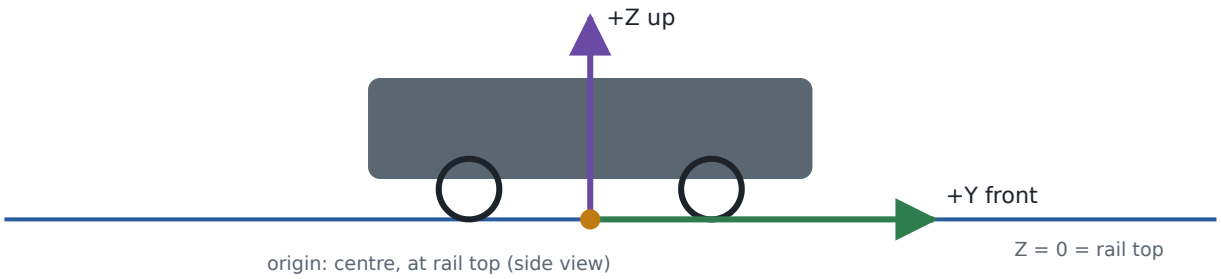


Figure 9.1 Colour follows the layer key: grey = model prep, amber = mechanism, violet = rig, blue = track, green = library.

9.2 Step by step

1 **Prepare the model**

- **Real-world scale in metres.** Check a known dimension (a mine cart's gauge is roughly 0.6 m; standard gauge is 1.435 m). Wheel rotation and cant maths depend on it.
- **Orientation:** front of the vehicle toward **+Y**, up **+Z**. This matches Follow Path's `FORWARD_Y / UP_Z`, which Track It always uses.
- **Apply** rotation and scale on every part (Ctrl+A). Leftover object scale breaks bone placement and wheel diameters.
- **Rail-top origin:** the model should sit so its wheel treads touch Z = 0. That plane is where the curve will be.



2 **Split into parts and set their origins**

Each thing that moves on its own must be its own object, with its **origin at its rotation centre**:

Part	Origin at	Name (convention)
Body / tub	vehicle centre, rail top	`MC_Body`
Each wheelset (or wheel)	axle centre	`MC_Wheel_F`, `MC_Wheel_R` (or `_F_L`, `_F_R`)
Frame / chassis	vehicle centre	`MC_Frame`
Couplers, doors, tipping hinge…	their hinge	`MC_Coupler_F`, `MC_Door`

Tip: in Edit Mode select the axle's edge loop, Shift+S → Cursor to Selected, then in Object Mode use Set Origin → Origin to 3D Cursor. These are the same manual steps the early sessions used.

3 Measure `Tool: Measure Vehicle`

With the parts named, a **Measure Vehicle** operator can read everything from bounding boxes: axle Y positions (from wheel origins), wheel diameters (bounding-box height), overall length, and coupler tips (the furthest ±Y points of the coupler objects). By hand, read each wheel's Location Y and Dimensions Z in the N-panel.

```
Mine cart (example)     axle F  +0.45 m   axle R  -0.45 m   wheel Ø 0.36 m
                        front coupler +0.95 m   rear coupler -0.95 m   length 1.90 m
```

4 Make Vehicle `Tool: wizard`

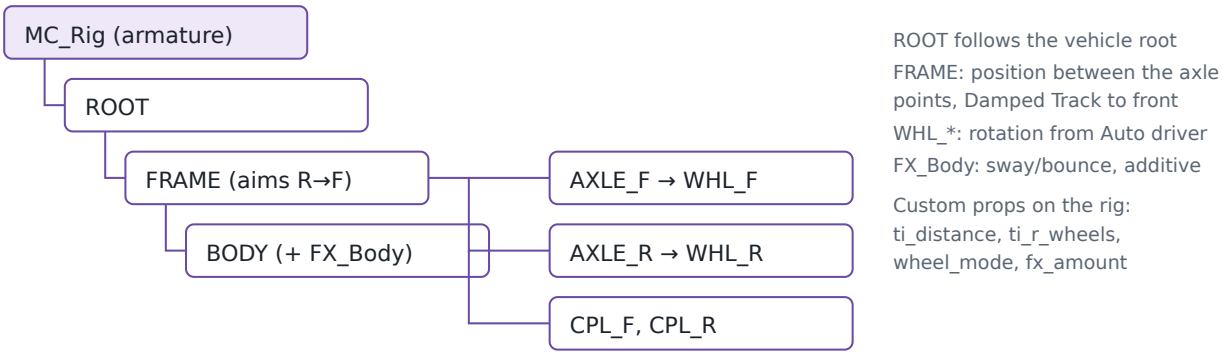
Select all the parts and run **Make Vehicle**, then pick a template:

Template	Creates
Rigid 2-axle (mine cart, wagon)	Root + 2 AXLE_POINT pivots at the measured axle offsets
Bogie car	Root + 2 TRACKED pivots at the bogie centres
Steam loco (Whyte n-n-n)	Root + drivers as AXLE_POINTs + RADIAL trucks (Part 10)
Custom	Root only. Add pivots by hand, as in today's Vehicle box

Today's equivalent is to add an Empty (or cylinder) as root at the vehicle origin, then use Vehicle → **+** twice with offsets **+0.45** and **−0.45**.

5 Generate the control rig `Tool`

The template builds an armature with standard bone names, placed at the part origins:



6 Bind the meshes to the rig

Parent each part to its bone (select part, Shift-select rig, Pose Mode, select bone, Ctrl+P → **Bone**). The tool does this automatically by name: `MC_Wheel_F` → `WHL_F`, and so on. Meshes **never** parent to the mechanism root or pivots (the “never bolted” rule).

7 **Wheel groups** Tool

Group wheels that share a diameter: here one group, “wheels”, Ø 0.36 m, mode AUTO. The tool adds the `d / r` driver (Part 7 §7.4) to each wheel bone's X rotation and the `grip` and `slip_angle` properties for later.

---

8 **Bind Rig**

Fill Pivot Data (today's stub): pivot 1 → `MC_Rig` / `AXLE_F`, pivot 2 → `AXLE_R`, then **Bind Rig**. This creates the `TI_` constraints of Part 6 §6.6. After binding, hide the pivot Empties.

---

9 **Test drive** Tool: test track

The add-on ships a test scene: an S-curve at the minimum radius, a banked curve, a turnout and a reverse loop. **Test Drive** runs the vehicle through it and reports:

- Pivots stay on the rails (distance to curve < 1 mm) and the wheelbase is held (±1 mm).
- Wheel travel matches the rotation: a marker on the tyre shouldn't slide along the rail.
- Body overhang on the minimum radius (this sets the clearance you need).
- No mesh parented to mechanism objects; no unapplied scale; all bones bound.

---

10 **Publish to the library**

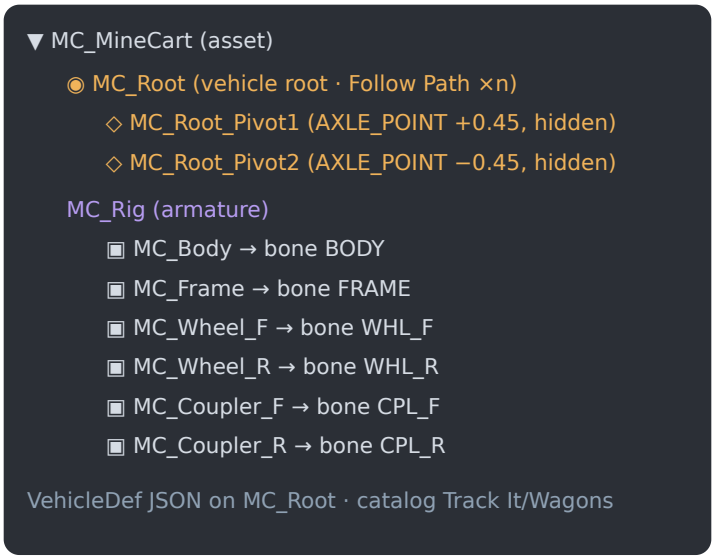
- Put root, pivots, rig and meshes in one collection, `MC_MineCart`. Right-click → **Mark as Asset**.
- Catalog: `Track It / Wagons`. Set a preview (render or auto).
- The Vehicle Definition (Part 6 §6.8) is stored on the root, so the asset carries everything.
- Save in a library .blend registered under Preferences → File Paths → Asset Libraries.

---

11 **Use it**

Drag the asset into a scene. The add-on detects a Track It vehicle and offers **Attach to nearest track** (the existing Attach logic). Add it to a consist from the Train Builder (**+ From Library**), then animate the consist's distance.

9.3 What the finished asset looks like



Checklist before publishing

- ☐ scale applied, metres, +Y forward
- ☐ wheels' origins on axle centres
- ☐ pivots match the measured axles
- ☐ every mesh parented to a rig bone
- ☐ no mesh parented to mechanism
- ☐ wheel groups + diameters set
- ☐ Bind Rig done, Empties hidden
- ☐ Test Drive passes
- ☐ couplers measured (front/rear)
- ☐ preview + catalog set

Track Doctor's vehicle checks automate most of this list.

Figure 9.2 Outliner view of the finished mine cart asset.

PROXY FIRST, FINAL MODEL LATER

Because meshes only parent to rig bones, you can publish a vehicle built from **proxy blocks** (as planned for the 2-4-2), animate whole scenes with it, and later swap in the finished model by re-parenting its parts to the same bones. The animation doesn't change at all.